

Species associations revealed by an automatic fauna identifier and 1.2 million citizen-science photographs

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Author: Gustavo Zafra · Creator and developer of BioFauna (BioCLIP-2.5 ViT-H/14 + FAISS identifier) and of the FotoFauna and BioQuest applications; contributor to the Minka SDG citizen-science platform.

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Abstract

Marine citizen science has produced photographic observation datasets that are rarely exploited beyond species distributions. We analyse **1,222,170 photographs** and **759,207 geolocated, dated observations** (Minka SDG, iNaturalist and other public sources), automatically identified by **BioFauna** —a k-NN retriever over BioCLIP-2.5 ViT-H/14 embeddings with hierarchical calibration— to uncover **species associations**. We define a sampling *event* (observer × ~1 km cell × day × 3-h window) and evaluate pair co-occurrence with support, spatial and temporal repeatability, finding **70,971 significant pairs**. The method ****recovers known, documented associations* with high robustness** —*e.g.* *Peltodoris atromaculata on the sponge Petrosia ficiformis** (79 events, lift 11.1, 44 localities, 74 days)— and reveals **undescribed candidates** with

plausible biological interpretation, such as jellyfish with commensal hyperiid amphipods, cleaner shrimps with anemones and infaunal species pairs. The work highlights the role of human curators in verifying cryptic species and outlines applications for conservation and outreach.

Keywords: citizen science, interspecific associations, habitat use, deep learning, Mediterranean, underwater photography.

1. Introduction

Amateur underwater photography has become a massive source of biodiversity data. Citizen-science platforms (**Minka SDG**, driven by the **FEC DAS** ecosystem; **iNaturalist**; **GBIF**) accumulate millions of verified observations, and automatic identifiers can consolidate that archive at catalogue scale. Yet most analyses stop at inventories and distribution maps. Photographs also carry **ecological context**: which species occur together, on what substrate and under which conditions. This work explores that latent information.

2. Material and methods

Summary (details in `METODOS.md` and supplementary material):

- **Data:** 1,222,170 photographs; 759,207 observations with coordinates, date and (92 %) time; a curated catalogue of 2,985 Mediterranean taxa.
- **Identification:** BioFauna (BioCLIP-2.5 ViT-H/14 + FAISS k-NN, k=15, T=0.05 aggregator with a cap of 3 votes per species, hierarchical calibration), with publication guards and leak auditing.
- **Unit of analysis:** *event* = (observer × ~1.1 km cell × day × 3-h window); the set of species seen together.
- **Filters:** catalogue-only; support ≥8 events; repeatability across ≥3 **localities**, ≥3 **days** and ≥5 **distinct observers** (the latter added after finding that a single dive trip generates false associations).
- **Metrics:** observed co-occurrence **n**, expected from marginals and **lift**; Poisson p for screening.
- **Internal validation:** blind search for documented associations (§3.1).

3. Results

3.1 Method validation (documented associations recovered)

association	n	lift	localities	days	source
<i>Peltodoris atromaculata</i> — <i>Petrosia ficiformis</i>	79	11.1	44	74	classic literature
<i>Felimare picta</i> — <i>Ircinia oros</i>	29	4.1	20	28	host sponge
<i>Cratena peregrina</i> — <i>Eudendrium racemosum</i>	5	10.5	—	—	prey hydroid
~~ <i>Doto paulinae</i> — <i>Sertularella mediterranea</i> ~~	5	48.3	—	—	✗ discarded : single observer (expert-review correction)

Co-occurrence is not an artefact of the confusion matrix (29-sep-2026): documented pairs show **null cross-confusion** (0-1 cases of $A \rightarrow B$ or $B \rightarrow A$ in 3-19 samples per species), while confusable sister-species pairs (control: *Elysia marginata/ornata*, *Spirobranchus lamarcki/triqueter*) show high cross-confusion (13-26 %) and do **not** appear as co-occurrence candidates. Observer bias is controlled by requiring ≥ 5 distinct observers. Detail: (detail in the supplementary material of the public repository, [papers/proyecto_correlacion/](#)).

3.2 Taxonomic groups

Pairs concentrate in the photographically best-represented families (sparids, blenniids, gobiids, labrids, coralline algae, didemnids, holothurians, actiniids, trochids, dictyotaceans, facelinids, serpulids, sabellids, irciniids). **98 % of pairs are inter-family**, as expected for habitat or trophic associations.

3.3 New candidates (pending expert review)

functional group	A — B	n	lift	localities	days
commensalism on jellyfish	<i>Blackfordia virginica</i> — <i>Phronima sedentaria</i>	19	653	6	15
cleaning symbiosis	<i>Lysmata grabhami</i> — <i>Telmatactis cricoides</i>	9	547	7	8
infauna (same sediment)	<i>Fustiaria rubescens</i> — <i>Loripinus fragilis</i>	16	892	7	15
infauna	<i>Abra alba</i> — <i>Abra longicallus</i>	9	534	7	9
hard-substrate microhabitat	<i>Macrorhynchia philippina</i> — <i>Telmatactis cricoides</i>	8	765	5	6
gelatinous plankton	<i>Callianira bialata</i> — <i>Vanadis formosa</i>	9	645	7	9

(Full table: supplementary material; main figure: top-12 network.)

3.4 Literature verification of the candidates

After reviewing public literature (WoRMS, Sea Slug Forum, FishBase, JNCC/MarLIN, peer-reviewed journals), pairs are classified into three categories:

Documented (the interaction is already described): *Peltodoris atromaculata*—*Petrosia ficiformis*

(predation), *Felimare picta*—*Ircinia* spp. (diet), *Cratena peregrina*—*Eudendrium racemosum*

(kleptopredation), the floating **Sargassum community** (*Latreutes fucorum*, *Hippolyte coerulescens* and *Scyllaea pelagica*), *Phronima sedentaria* as a salp symbiont and *Lampea pancerina* as a salp predator/parasite.

Plausible (ecologically coherent, no direct citation of the pair): *Lysmata grabhami*—**Telmatactis cricoides* (*the anemone hosts documented crustacean symbionts* —Thor amboinensis* in 65 % of anemones and *Stenorhynchus lanceolatus* in facultative mutualism— and both species truly co-occur in the Canary Islands/Madeira), gelatinous-plankton pairs (ctenophores, hyperiids and medusae of the same water mass) and infaunal sediment pairs (*Abra* spp.; *Fustiaria*—*Loripinus*, both typical of circalittoral soft bottoms).

Dive-trip artefact (species from other oceans photographed together by the same group in one immersion):

Doris fontainii—*Tyrinna delicata* (both from the **South Pacific**, lat −12...−55),

Chromodoris quadricolor—*Hexabranhus sanguineus* (Red Sea/Indo-Pacific), *Abudefduf saxatilis*—

Kyphosus vaigiensis (Atlantic vs. Indo-Pacific; they do not coexist in the Mediterranean). These pairs are

excluded from the ecological analysis of the study area; detecting them is also a useful dataset-quality control signal.

Taxonomic correction: `loripinus_fragilis` (Philippi, 1836, **Lucinidae**) and `limaria_fragilis` (Gmelin, 1791, **Limidae**) are distinct species; they are not synonyms (WoRMS AphiaID 718970 vs. 216644).

3.5 References supporting the associations

Sources verified during the literature review (28-sep-2026):

Association / claim	Source
<i>Peltodoris atromaculata</i> feeds on the sponge <i>Petrosia ficiformis</i>	trophic literature of dorid nudibranchs (diet review; petroformynes as chemical defence of the sponge)
<i>Felimare picta</i> (formerly <i>Hypselodoris</i>) feeds on sponges of the genus <i>Ircinia</i>	McDonald & Nybakken, 2001 — <i>A worldwide review of the food of nudibranch mollusks</i> ; field observations (OPK)
<i>Cratena peregrina</i> practises kleptopredation on <i>Eudendrium racemosum</i>	Di Camillo et al. — <i>Eudendrium racemosum</i> as substrate/prey and prey theft (CNR, Adriatic Sea)

> community.

<i>Scyllaea pelagica</i>, <i>Latreutes fucorum</i> and <i>Hippolyte coerulescens</i> belong to the floating <i>Sargassum</i> fauna	literature of the pelagic <i>Sargassum</i> community (NOAA; BAMZ; Sea Slug Forum)
<i>Phronima sedentaria</i> is an obligate salp symbiont (lives in the "barrel")	Laval, 1978 — <i>The barrel of the pelagic amphipod Phronima sedentaria</i> ; Diebel, 1988 — <i>Observations on the anatomy and behavior of Phronima sedentaria</i> ; MBARI/Scripps
<i>Lampea pancerina</i> predaes or parasitises salps	<i>The double life of the ctenophore Lampea pancerina</i> ; ctenophore literature (Carré & Carré)
Hyperiid (<i>Brachyscelus</i> , <i>Phronima</i>) associate with gelatinous zooplankton (medusae, ctenophores, salps)	Keil & Osborn — <i>Associations between hyperiid amphipods and gelatinous zooplankton</i> (Smithsonian/MBARI, ROV)
<i>Telmatactis cricoides</i> hosts crustacean symbionts (<i>Thor amboinensis</i> in 65 % of anemones; <i>Stenorhynchus lanceolatus</i> in facultative mutualism)	<i>Crustacean symbionts of the sea anemone Telmatactis</i> (decapoda.nhm.org); Peraza et al., 2024 — <i>Exploring the association between Stenorhynchus lanceolatus and Telmatactis cricoides in the Canary Islands</i> (Regional Studies in Marine Science)
<i>Telmatactis cricoides</i> is a thermophilic species expanding in the western Mediterranean (Almería, Balearic Is.)	Cambridge, J. Mar. Biol. Assoc. UK 104 (2024) — <i>The thermophilic sea anemone Telmatactis cricoides in the western Mediterranean</i>
<i>Macrorhynchia philippina</i> is an invasive hydrozoan ("stinging bush hydroid")	Riera et al., 2016 — <i>Progressing the invasion of the hydrozoan Macrorhynchia philippina in Atlantic archipelagos</i> (Vieraea 44)
Soft-sediment communities with <i>Abra alba</i> and same-genus bivalves	JNCC Marine Habitat Classification (biotope SS.SSa.CMuSa.AalbNuc); MarLIN
<i>Loripinus fragilis</i> (Lucinidae, chemosymbiosis) ≠ <i>Limaria fragilis</i> (Limidae)	WoRMS AphiaID 718970 vs. 216644; Taylor & Glover, 2021 — <i>Biology, evolution and generic review of the Lucinidae</i>
<i>Abudefduf saxatilis</i> is strictly Atlantic (replaced by <i>A. vaigiensis</i> in the Indo-Pacific)	FishBase — <i>Abudefduf saxatilis</i> (Sergeant-major)
<i>Doris fontainii</i> (formerly <i>Anisodoris fontainei</i>) and <i>Tyrinna delicata</i> are South Pacific species	Sea Slug Forum (Australian Museum); Valdés & Muniain, 2002
<i>Eurythoe complanata</i> is a complex of ≥3 cryptic species	Barroso et al., 2010 — <i>Eurythoe complanata, the 'cosmopolitan' fireworm, consists of at least three cryptic species</i> (Marine Biology)

3.6 Trophic associations of nudibranchs (29-sep-2026)

Nudibranchs are an ideal group to validate the method: their diet is **specialist and well documented** (each species eats one or few colonial prey —sponges, hydroids, bryozoans, ascidians, algae—; Wägele & Klussmann-Kolb, 2005; McDonald & Nybakken, 2001; Sea Slug Forum). Of the **70,971** robust pairs, ****9,539** (13.4 %) involve at least one nudibranch, **and** 1,048* *are* nudibranch → possible prey/substrate* pairs with **≥5 distinct observers** (sponges, hydroids, bryozoans, ascidians and algae of the catalogue). The method **recovers published diets** as the strongest associations:

Nudibranch (family)	Co-occurring prey/substrate	n	obs.	lift	cells	bibliographic source
<i>Doto floridicola</i> (Dotidae)	<i>Aglaophenia elongata</i> (hydroid)	9	5	59.1	7	✓ documented: <i>Doto</i> spp. feed on <i>Aglaophenia</i> hydroids (Pictou, Sea Slug Forum)
<i>Felimare orsinii</i> (Chromodorididae)	<i>Scalarispongia scalaris</i> (sponge)	12	5	65.3	10	✓ documented: "feeds primarily on the sponge <i>Scalarispongia scalaris</i> " (Sea Slug Forum; SEASLUG.WORLD)
<i>Trinchesia caerulea</i> (Trinchesiidae)	<i>Sertularella crassicaulis</i> (hydroid)	19	6	83.4	15	✓ coherent: aeolids eat sertulariid hydroids
<i>Trinchesia caerulea</i>	<i>Turbicellepora avicularis</i> (bryozoan)	17	5	44.6	14	coherent (bryozoans as secondary prey/substrate)
<i>Aeolidiella alderi</i> (Aeolidiidae)	<i>Tedania anhelans</i> (sponge)	14	6	74.2	6	coherent (aeolids on sponges/hydroids)
<i>Dendrodoris limbata</i> (Dendrodorididae)	<i>Aplidium turbinatum</i> (ascidian)	23	9	62.6	8	coherent (dorids on colonial ascidians)
<i>Siphonaria pectinata</i> (Siphonariidae)	<i>Bifurcaria bifurcata</i> (brown alga)	26	14	61.1	16	✓ coherent: <i>Siphonaria</i> are herbivorous pulmonate limpets
<i>Taringa armata</i> (Discodorididae)	<i>Aplysilla sulfurea</i> (sponge)	12	4	89.4	4	coherent (sponge-eating dorids)
<i>Haminoea navicula</i> (Haminoeidae)	<i>Aplidium turbinatum</i> (ascidian)	14	6	186.2	3	coherent (cephalaspideans on hard substrate)

The agreement between measured co-occurrence and published diets is ****the strongest external validation of the method: it not only recovers documented pairs from the classic literature, but** ranks first by lift** the cases with known diet (*Doto* → *Aglaophenia*, *Felimare orsinii* → *Scalarispongia*). The rest of the list (1,039 candidates) constitutes testable diet/substrate hypotheses for understudied Mediterranean nudibranchs.

Artefact: `artifacts/coocurrencia_nudibranquios_trofica_filtrada_20260929.json` (1,048 pairs with ≥5 observers).

3.7 Additional relationships beyond nudibranchs (29-sep-2026)

The same analysis applied to the **whole catalogue** (not only nudibranchs) with an anti-"hub" filter (the 39

species co-occurring with >400 taxa are excluded as background noise) yields **11,509 specific Mediterranean pairs** with ≥ 5 observers. Three groups of relationships stand out with ecological and conservation value:

a) Co-occurrence of Lessepsian invaders (Red Sea species that entered via the Suez Canal):

A	B	n	obs.	lift	context
<i>Fistularia commersonii</i> (cornetfish)	<i>Pterois miles</i> (lionfish)	25	16	261.7	both documented invaders; <i>Fistularia</i> is one of the few natural predators of <i>Pterois</i>
<i>Fistularia commersonii</i>	<i>Siganus rivulatus</i> (rabbitfish)	15	13	123.9	coexisting invaders in the Levant
<i>Pterois miles</i>	<i>Taeniura lymma</i> (bluespotted ray)	11	7	233.2	Red Sea invaders

The co-occurrence of Lessepsian invaders is **consistent with the literature** (Kondylatos et al., 2023):

Fistularia, *Pterois miles* and *Siganus* caught together in Rhodes; CIESM Atlas of Exotic Fishes).

Conservation implication: citizen-science photographs allow **monitoring the joint spread** of these invasive species.

b) Gelatinous plankton (ctenophores, siphonophores and their predators/commensals):

A	B	n	obs.	lift
<i>Forskalia edwardsii</i> (siphonophore)	<i>Lampea pancerina</i> (ctenophore)	8	5	209.9
<i>Cestum veneris</i> (Venus' girdle)	<i>Hippopodius hippopus</i> (siphonophore)	18	7	102.6
<i>Callianira bialata</i> (ctenophore)	<i>Cestum veneris</i>	16	5	97.7
<i>Callianira bialata</i>	<i>Forskalia edwardsii</i>	14	5	177.1

Ctenophores prey on salps and gelatinous zooplankton (Carré & Carré); their co-occurrence with siphonophores

in the same water mass is coherent with a **shared planktonic predatory guild** (Current Biology, 2025).

c) Benthic invertebrates (habitat associations with biological meaning):

A	B	n	obs.	lift
<i>Astroides calycularis</i> (orange coral)	<i>Clavelina dellavallei</i> (ascidian)	16	8	135.6
<i>Aeolidiella alderi</i> (nudibranch)	<i>Berthella perforata</i> (pleurobranch)	41	8	129.8
<i>Petalifera petalifera</i> (sea hare)	<i>Placida tardyi</i> (sacoglossan)	22	5	110.7
<i>Codium coralloides</i> (green alga)	<i>Placida verticilata</i> (sacoglossan)	18	5	101.0
<i>Oostergrenia digitata</i> (sea cucumber)	<i>Virgularia mirabilis</i> (sea pen)	8	5	212.9

The pair *Codium coralloides*—*Placida verticilata* is especially notable: **sacoglossans feed on green algae of the genus *Codium*** (retaining their chloroplasts; Wägele & Klussmann-Kolb, 2005) — a documented trophic relationship the method recovers without prior knowledge.

4. Discussion

4.1 Theoretical background: co-occurrence and citizen-science biases

Species co-occurrence is a classic ecological signal (shared habitat, predation, symbiosis, commensalism), but reading it from citizen-science data requires controlling well-documented biases: **geographic bias** (accessible areas are sampled more), **effort bias** (bioblitz-type events inflate certain localities), **observer bias** (variable ability to detect and identify species) and **reporting bias** (preference for charismatic species). Opportunistic data lack a sampling plan and explicit absences, so any inference must explicitly mitigate these biases (Isaac et al., 2014; Johnston et al., 2018; Aceves-Bueno et al., 2017; Boyd et al., 2021; Milanesi et al., 2020). Our design responds with: (1) *event* = effort unit per observer and 3-h window (controls effort bias), (2) required repeatability across ≥ 3 localities, ≥ 3 days and ≥ 5 distinct observers (**controls observer and dive-trip bias**), and (3) the geographic-range filter** on species (removes the dive-trip artefact found in this review). Geographic bias is addressed with the locality·date stratified null currently under development (§6 of Methods).

Candidates fall into three interpretative classes: **commensalism/phoresy** (jellyfish—amphipod), **cleaning symbiosis (**shrimp—anemone**) and substrate co-habitation** (infauna; hard bottoms; coralline algae). The central methodological challenge is separating **interaction** from **coincidence** driven by habitat, season or observer effort; hence the spatial/temporal repeatability requirement and the stratified null currently under development.

The literature review yields two results of general interest. First, **the method recovers already documented associations** (doris predation, the Sargassum community, gelatinous-zooplankton symbionts), validating the ecological signal against effort noise. Second, classifying the top pairs by the geographic range of their species reveals a **systematic citizen-science artefact**: the highest-lift pairs with extra-Mediterranean species (*Doris fontainii*—*Tyrinna delicata*; *Chromodoris quadricolor*—*Hexabranhus sanguineus*) correspond to **dive trips** by the same group rather than to ecological interactions in the study area. Cross-checking each taxon's geographic range is therefore a necessary quality filter before interpreting any candidate as an ecological association.

4.2 Contrasted biological hypotheses (the main scientific contribution)

The robust candidates are here confronted with the marine biological literature. Each case is classified as

documented (the interaction is already described), **plausible** (ecologically coherent, no direct citation of the pair) or **artefact** (sampling bias, discarded).

(a) *Peltodoris atromaculata* — *Petrosia ficiformis* (predation). Documented. The dorid feeds exclusively on that sponge, accumulates its petroformynes and lives on it; the method recovers it with 79 events, 44 localities and 74 days (lift 11.1). *Doto floridicola*—*Aglaophenia* and *Felimare orsinii*—**Scalarispongia scalaris** confirm the general pattern: **dorids and dotids co-occur with their prey sponge/hydroid** (Sea Slug Forum; McDonald & Nybakken, 2001).

(b) *Cratena peregrina* — *Eudendrium racemosum* (kleptopredation). Documented (CNR, Adriatic Sea): the aeolid steals prey from the hydroid. Our pair (n=5, 5 observers) is consistent with the literature.

(c) Floating *Sargassum* community (*Scyllaea pelagica*, *Latreutes fucorum*, *Hippolyte coerulea*).

Documented: the three species are fauna associated with the floating alga *Sargassum* (NOAA; BAMZ). The method groups them with 10-11 distinct observers and lift >300, without knowing the relationship a priori.

(d) *Lysmata grabhami* — *Telmatactis cricoides* (cleaning/symbiosis). Plausible. The anemone **T. cricoides* hosts documented crustacean symbionts (*Thor amboinensis* in 65 % of anemones; *Stenorhynchus lanceolatus* in facultative mutualism — Peraza et al., 2024); *Lysmata** are cleaner shrimps. They truly co-occur in the Canary Islands/Madeira (lat 27-29). Requires in-situ validation.

(e) *Blackfordia virginica* — *Phronima sedentaria* / *Brachyscelus cruscum* (commensalism on gelatinous zooplankton). Plausible. *Hyperiid*s associate with *medusae* and *salps* (Keil & Osborn; *Phronima** is an obligate salp symbiont — Laval, 1978). No specific *Blackfordia*-hyperiid citation; testable hypothesis.

(f) Infaunal pairs (*Fustiaria rubescens*—*Loripinus fragilis*; *Abra alba*—*Abra longicollis*).

Plausible. Co-habitation of circalittoral soft bottoms (JNCC biotope "Abra alba & *Nucula nitidosa*"; *Loripinus* lives in reduced sediments with chemosynthetic symbionts).

Discarded as artefacts: the highest-lift pairs with extra-Mediterranean species (*Doris fontainii*—*Tyrinna delicata*, South Pacific; *Chromodoris quadricolor*—*Hexabranhus sanguineus*, Red Sea) are dive-trip photos of the same group, not associations of the study area; detecting them is also a dataset-quality control signal.

Implication: five hypotheses (d, e, f and variants) require **in-situ validation** by biologists — targeted dives, fixed cameras or review of pairing photographs. That is the line of work that turns the result into a citable scientific contribution and the basis of the BioQuest application ("if you see X, look for Y").

5. Limitations

1. Co-occurrence $\neq$ interaction; mitigated by filters and the null model, but directed observation remains the final confirmation.
2. Observer/effort bias (bioblitz-type events).
3. Uneven taxonomic coverage.

6. Conclusions and applications

Citizen-science photographs contain ecological associations retrievable at catalogue scale. Results enable (a) **prioritising** the search for rare species where their associate occurs, (b) **enriching BioQuest** with "if you see X, look for Y" suggestions and per-area association networks, and (c) generating ****testable hypotheses**** for marine research, with the added value of covering shallow coastal waters where satellites perform worst.

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Supplementary material

- Association tables (top 40 and full set; available in the project's public repository).
- Association network figure (navigable SVG in the repository).
- **Photo plates** (LAMINAS_v4_20260929 . pdf , 22 pages): 22 **photographs** selected from the ****BioFauna gallery (1.2 M images) with the** system quality classifier** (**obs_score** : research grade, number of curators, trusted curator and resolution; 0-13) and a sharpness filter (Laplacian variance). Each photo includes its **attribution** (author + licence + observation link). Only reusable-licence images (CC0 / CC BY / CC BY-SA); the rest are excluded or require the author's explicit permission.